7.5.5 Types of capacitive transducers

a. Capacitive transducers - By variation overlapping area

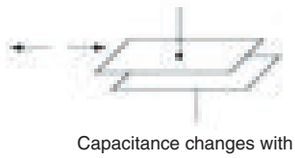
This type of transducers operates on the fact that capacitance of the capacitor is proportional to the overlapping area of plates. As shown in the figure 7.12, one plate is fixed and the other is movable. If the displacement is taken place, one of the plates connected to displacement moves over the fixed plate, thereby changing the overlapping area.

b. Capacitive transducers - By variation of distance between the plates

This type of transducers operates on the fact that capacitance of the capacitor is inversely proportional to the distance between the plates. Such type of transducers are used to measure the linear displacement as shown in figure 7.12.

c. Capacitive transducers - By variation of permittivity of the di-electric material

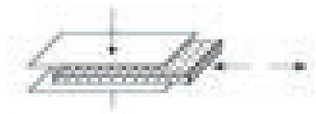
This type of transducers operates on the fact that capacitance of the capacitor is varied by varying the permittivity of the dielectric material. In this arrangement, a dielectric material of permittivity 'e' moves between the two fixed plates as shown in figure 7.12.



Capacitance changes with changes in plate overlap



Capacitance changes with changes in distance



Capacitance changes with changes in dielectric

Fig. 7.12 Capacitive transducers

d. Advantages of capacitive transducers

i. Good frequency response.
ii. These are not affected by stray magnetic fields.
iii. These are extremely very sensitive.
iv. Operates at a very little force.

e. Disadvantages

i. The output changes with temperature.
ii. Adequate design is needed for accurate measurements.
iii. Electrostatic screening is needed.

7.6 Piezo electric transducers

The piezo electric sensors are based on the principles of electro mechanical energy conversion. The mechanical input is converted to the electrical output and that is the basis of this transducers and this transducer shows piezo electric effect.

In some crystalline materials, a potential is developed across the opposite faces of the material when a mechanical force is applied on it. This is called piezo electric effect and the materials which exhibit this behavior is called as piezo electric materials. This effect is used to measure the dynamic pressure, force, shock and vibratory motion. Common piezo-electric materials are rochelle salt, ammonium de-hydrogen phosphate, lithium sulphate and quartz ceramic.

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